

22PH202 – Electromagnetism and Modern Physics\

2.19 Classification of magnetic materials

The origin of magnetism lies in the orbital and spin motions of electrons and how the electrons interact with one another. The best way to introduce the different types of magnetism is to describe how materials respond to magnetic fields. This may be surprising to some, but all matter is magnetic. It's just that some materials are much more magnetic than others. The main distinction is that in some materials there is no collective interaction of atomic magnetic moments, whereas in other materials there is a very strong interaction between atomic moments.

The magnetic behavior of materials can be classified into the following three major groups:

- Diamagnetism
- Paramagnetism
- Ferromagnetism

2.19.1 Diamagnetic materials

These materials are barely magnetised when placed in a magnetic field. Magnetic dipoles in these substances tend to align in opposition to the applied field. In effect, they produce an internal magnetic field that opposes the applied field and the substance tends to repel the external field around it.

This opposing field disappears as soon as the external field is removed.

Ex: Gold, water

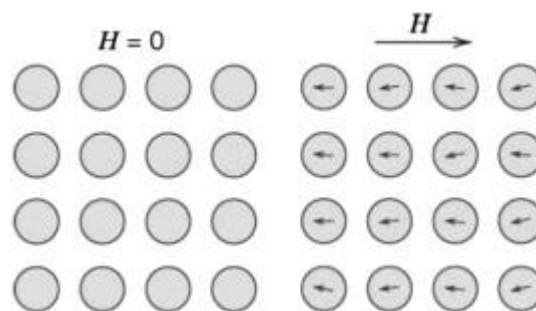


Figure 37. Magnetic moment of diamagnetic materials

- No permanent dipoles.
- The induced magnetic moment in the diamagnetic material is always opposite to the direction of applied field.

- Repels magnetic lines of force.
- Magnetic susceptibility is negative and is independent of temperature and applied field.
- Permeability is less than 1.

2.19.2 Paramagnetic materials

In these materials, the magnetic dipoles in the magnetic materials tend to align along the applied magnetic field and thus reinforcing the applied magnetic field. Such substances are attracted by a magnet if it applies a sufficiently strong field. It must be noted that such materials are still feeble magnetised and the magnetisation disappears as soon as the external field is removed.

Ex: Liquid oxygen, sodium, platinum, salts of iron and nickel.

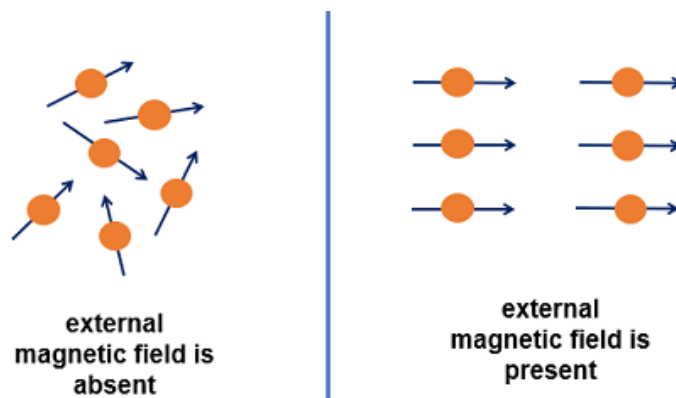


Figure 38. Magnetic moment of paramagnetic materials

- It has permanent dipoles.
- The induced magnetic moment is always same as the direction of applied field.
- Paramagnetic materials are attracted by magnets
- Magnetic susceptibility is very small and positive.
- Permeability is slightly greater than 1.
- Susceptibility decreases with increase in temperature as given by Curie –Weiss Law.

2.19.3 Ferromagnetic materials

We are most familiar with these materials as they exhibit the strongest magnetic behaviour. Magnetic dipoles in these materials are arranged into domains where the arrangements of individual magnetic dipoles are essentially perfect that can produce strong magnetic fields. Normally, these domains are usually randomly arranged and thus the magnetic field of each domain is cancelled by another and the entire material does not show any magnetic behaviour.

However, when an external field is applied, the domains reorient themselves to reinforce the external field and produce a strong internal magnetic field that is along the external field. Upon removal of the external field, most of the domains stay put and continue to be aligned in the direction of the (erstwhile) magnetic field. Thus, the magnetic field of the magnetic materials persists even when the external field disappears. This property is used to produce Permanent magnets that we use every day. Iron, cobalt, nickel, neodymium and their alloys are usually highly ferromagnetic and are used to make permanent magnets.

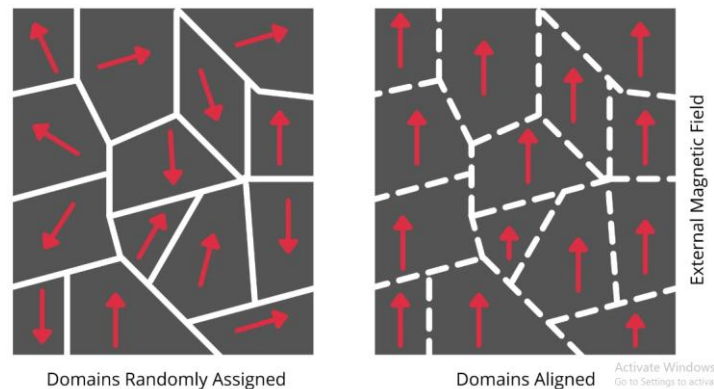


Figure 39. Magnetic moment of Ferromagnetic materials

- High degree of magnetization due to spin exchange interaction.
- Strongly attracted by magnets
- Magnetic susceptibility is positive and large.
- Permeability is very large.
- Placed in magnetic field – easily magnetized.
- Susceptibility decreases with increase in temperature as given by Curie –Weiss Law.
- When the temperature is greater than curie temperature, the ferromagnetic material is converted into paramagnetic material.